Inequalities: Using Graphs

Questions 1–11

Question 1. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x-3}{x+2}$$

a) State the asymptotes of the curve.
b) Sketch $y = \mathrm{f}(x)$
c) Hence, giving your answer in set notation, find the values when

$$\frac{x-3}{x+2} < 0$$

Question 2. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x+4}{x-1}$$

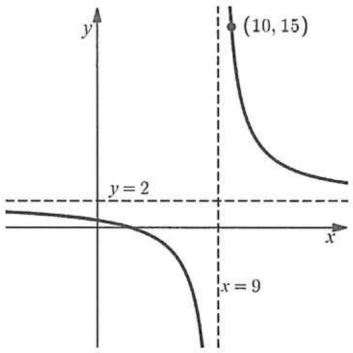
a) State the asymptotes of the curve.
b) Sketch $y = \mathrm{f}(x)$
c) Hence, giving your answer in interval notation, find the values when

$$\frac{3x+4}{x-1} \geq 0$$

Question 3. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{2x - a}{x - b}$$

where a and b are positive integers is shown in the diagram below.



The asymptotes of the curve are given by the equations $x = 9$ and $y = 2$

a) State the values of a and b

b) Find the coordinates of where the curve meets the x-axis and y-axis.

c) Hence, find the values when

$$\frac{2x - a}{x - b} > 0$$

Give your answer in set notation.

Question 4. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{1 - 3x}{4x + 5}$$

a) State the asymptotes of the curve.

b) Sketch $y = \mathrm{f}(x)$

c) Hence, find the values when

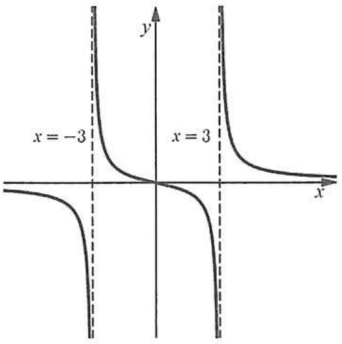
$$\frac{1 - 3x}{4x + 5} < \frac{1}{2}$$

Give your answer in set notation.

Question 5. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x}{x^2 - k}$$

where k is a positive integer is shown in the diagram below.



The asymptotes of the curve are given by the equations $x = -3$, $x = 3$ and $y = 0$

a) State the value of k

b) For your value of k, solve

$$\frac{x}{x^2 - k} = \frac{2}{5}$$

c) Hence, for your value of k find the values when

$$\frac{x}{x^2 - k} \geq \frac{2}{5}$$

Give your answer in set notation.

Question 6. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 + x - 20}{x^2 + x - 6}$$

a) State the asymptotes of the curve.

b) Sketch $y = \mathrm{f}(x)$

c) Hence, find the values when

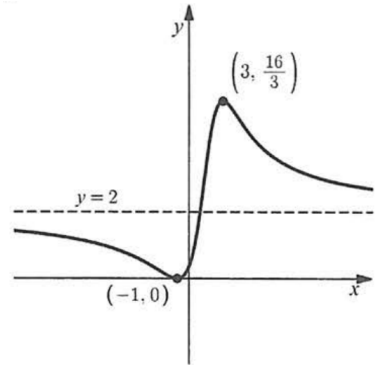
$$\frac{x^2 + x - 20}{x^2 + x - 6} \leq \frac{9}{2}$$

Give your answer in interval notation.

Question 7. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + 4x + b}{x^2 - 3x + c}$$

where a, b and c are integers is shown in the diagram below.



The asymptote of the curve is $y = 2$. There is a maximum at $(3, \frac{16}{3})$ and a minimum at $(-1, 0)$

a) Find the values of a, b and c

b) Hence, find the values when $\mathrm{f}(x) < 2$
Give your answer in interval notation.

Question 8. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 - 1}{2x^2 - 4x + 3}$$

a) State any asymptotes of the curve.

The line $y = k$ intersects the curve.

b) Show that $(2k + 1)(k - 1) \leq 0$

c) Hence find the coordinates of the stationary points of the curve and identify their types.

d) Hence sketch $y = \mathrm{f}(x)$

e) Find the values of x when

$$\frac{x^2 - 1}{2x^2 - 4x + 3} \geq x - 1$$

Give your answer in set notation.

Question 9. It is given that a rational function has equation $y = \mathrm{f}(x)$. Its graph has exactly two asymptotes $y = -2$ and $x = 1$
It intersects the x-axis exactly once at $(3, 0)$

a) Find $\mathrm{f}(x)$
b) Hence state the coordinates where the curve meets the x-axis and the y-axis.
c) Sketch $y = \mathrm{f}(x)$
d) Hence, find the value(s) for which $\mathrm{f}(x) \geq 4$
Give your answer in interval notation.

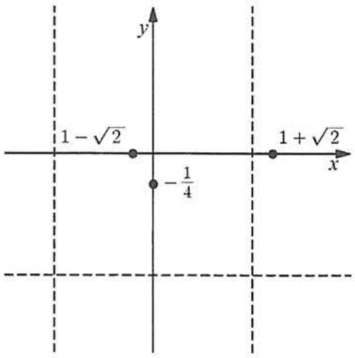
Question 10. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x^2 + ax + b}{x^2 + cx + 2}$$

where a, b and c are integers.
It has exactly one asymptote with the equation $y = 3$. The curve passes through the y-axis at -1. The curve has a maximum at $(3, \frac{29}{7})$ and a minimum at $(-1, -5)$

a) Find $\mathrm{f}(x)$
b) Find the values of x where the curve meets the x-axis.
c) Sketch $y = \mathrm{f}(x)$
d) Find the value(s) for which $\mathrm{f}(x) \geq -1$
Give your answer in set notation.

Question 11. It is given that $y = \mathrm{f}(x)$ is a rational function of quadratic order in both the numerator and denominator of the fraction.
An incomplete sketch is shown below based on the following information:
Its graph has exactly three asymptotes which are given by equations $y = -1$, $x = 2$ and $x = -2$
It crosses the x-axis at $1 + \sqrt{2}$ and $1 - \sqrt{2}$ and it crosses the y-axis at $-\frac{1}{4}$



a) On a copy of the diagram above, complete the sketch of the curve.
You should label the asymptotes.
b) Find $\mathrm{f}(x)$
c) Hence, solve $\mathrm{f}(x) \leq -1$
Give your answer in interval notation.